

Skills Summary

One Line, Four Ways to Show It

Use this reference while completing your worksheet or when studying.

Skill 1 — Table to equation

Rule: the row where the input is 0 gives b ; the change in output divided by the change in input gives m . Then write $y = mx + b$.

Example: Write the equation for the table $x = 0, 2, 4$ with $y = 5, 13, 21$.

Step 1. The table already contains the input 0, so b can be read off and only m needs computing.

Step 2. At $x = 0$ the output is 5. Across the table, $\frac{21 - 5}{4 - 0} = \frac{16}{4}$.
Slope in front, intercept behind. **$y = 4x + 5$**

Try it: A table gives $x = 0, 3, 6$ with $y = 7, 16, 25$. Write its equation.

check: $7 + x8 = 16$

Skill 2 — Context to equation

Rule: the amount already there before anything happens is b ; the amount added (or lost) per unit is m . Words like “per”, “each” and “every” point at the slope.

Example: A club has 25 members and gains 9 more each month. How many after 5 months?

Step 1. One number is a starting amount and the other is a rate, so sort them before writing anything: 25 happens once, 9 happens every month.

Step 2. $y = 9x + 25$, and at $x = 5$ this gives $45 + 25$.

The rate multiplies the months. **70**

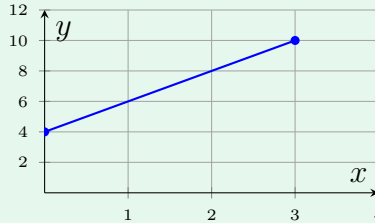
Try it: A tank holds 40 litres and gains 6 litres a minute. How many litres after 8 minutes?

check: 88

Skill 3 — Graph to equation

Rule: read where the line crosses the vertical axis for b , then count rise over run between two grid corners for m .

Example: Write the equation of the line drawn below, then find y at $x = 5$.



Step 1. The crossing point is marked on the axis, so take b from there and get m from the two marked corners.

Step 2. Crossing at 4; rise over run is $\frac{10 - 4}{3 - 0} = \frac{6}{3}$, so the equation is $y = 2x + 4$. At $x = 5$: $10 + 4$.

Reading past the edge of the grid needs the equation. $y = 14$

Try it: A line is marked at $(0, 1)$ and $(2, 7)$. Find its slope.

check: $m = 3$

Skill 4 — Deciding whether two forms match

Rule: two representations match only when *both* numbers agree. One shared value proves nothing — two different lines can cross.

Example: Does the table $x = 0, 2, 4$ with $y = 8, 14, 20$ match $y = 3x + 8$?

Step 1. Comparing one value could be a coincidence, so test the pair of numbers that defines a line: the starting value and the rate.

Step 2. Table at $x = 4$ gives 20; the equation gives $12 + 8$, the same. Both start at 8 and both climb 6 over 2 steps.

Agreeing on both, so yes. **20**

Try it: Does the table $x = 0, 3, 6$ with $y = 10, 19, 28$ match $y = 3x + 10$? Check the value at $x = 6$.

check: 28, and yes